The 60-Period Overlapping Generations Model

Auerbach–Kotlikoff Framework

Introduction

Historical Background

Building on the two-period OLG tradition:

- Allais (1947), Samuelson (1958), Diamond (1965) establish the two-period OLG framework.
- Auerbach and Kotlikoff (1987) extend it to **55 cohorts** (years), enabling realistic life-cycle analysis.
- Main use: quantitative evaluation of **fiscal policy** (tax reform, social security, public debt).

This Chapter

We study a **simplified 60-period version** and implement it via two computational strategies:

1. **Direct computation** – reduce FOCs to a nonlinear system and apply Newton-type methods.
2. **Value function iteration** – dynamic programming on a discrete asset grid.

The Model

Household Setup

- Each year, a new generation of **equal measure** is born.
- Population is constant and normalised: $N_t \equiv 1$.
- Agents live for $T = T^W + T^R = 40 + 20 = 60$ years; each cohort has mass $1/60$.
- Workers supply labour l_t^s at age s in period t, enjoying leisure $1 - l_t^s$.
- **Mandatory retirement** after age $T^W = 40$: $l_t^s = 0$ for $s > T^W$.

Lifetime utility – maximised at birth (age 1)

$$\sum_{s=1}^{T} \beta^{s-1} \, u\left(c_{t+s-1}^s, \; 1 - l_{t+s-1}^s\right)$$

Preferences

Instantaneous utility (CES–Stone–Geary form)

$$u(c, 1-l) = \frac{\left[(c+\psi)(1-l)^{\gamma}\right]^{1-\sigma} - 1}{1-\sigma}$$

Parameter	Interpretation
$\sigma > 0$	Inverse of the elasticity of intertemporal substitution (IES)
$\beta \in (0, 1)$	Time discount factor
$\gamma > 0$	Weight of leisure in utility
$\psi = 0.001$	Small constant ensuring u is finite at $c = 0$

Note: ψ is a pure **computational device**; it makes preference to be not homothetic.

Preferences

Instantaneous utility (CES–Stone–Geary form)

$$u(c, 1-l) = \frac{\left[(c+\psi)(1-l)^{\gamma}\right]^{1-\sigma} - 1}{1-\sigma}$$

Special case	Utility	Family
$\psi = 0,\ \sigma = 1$	$\log c + \gamma \log(1-l)$	King–Plosser–Rebelo
$\psi = 0$	$\dfrac{\left[c(1-l)^{\gamma}\right]^{1-\sigma} - 1}{1-\sigma}$	Standard CES-composite
$\sigma = 1$	$\log(c+\psi) + \gamma \log(1-l)$	Stone–Geary log
$\psi = 0,\ \gamma = 0$	$\dfrac{c^{1-\sigma} - 1}{1-\sigma}$	Pure CRRA (no leisure)

CES–Stone–Geary Utility: Main Properties

Instantaneous utility

$$u(c, 1-l) = \frac{\left[(c+\psi)(1-l)^{\gamma}\right]^{1-\sigma} - 1}{1-\sigma}, \qquad \tilde{c} \equiv (c+\psi)(1-l)^{\gamma}$$

Homotheticity

- $\text{MRS} = \dfrac{\gamma(c+\psi)}{1-l}$
- **Not homothetic** when $\psi \neq 0$: $\text{MRS}(\lambda c, \lambda h) \neq \text{MRS}(c, h)$
- Homothetic only if $\psi = 0$

Wealth effect on labour supply

- Static optimum: $1 - l = \dfrac{\gamma(c+\psi)}{w}$
- Higher $c \Rightarrow$ higher demand for leisure $\Rightarrow$ **wealth effect present**
- Contrast: GHH preferences eliminate it

Frisch elasticity of leisure

$$\varepsilon_F^{\text{leisure}} = \frac{\partial \log(1-l)}{\partial \log w} = \frac{\sigma}{\sigma(1+\gamma) - \gamma}$$

Jaimovich–Rebelo (2009): A Unified Family

Functional form.

$$U = \frac{1}{1-\sigma}\left[c - \psi\,\frac{l^{1+1/\eta}}{1+1/\eta}\,X_t\right]^{1-\sigma}$$

$$X_t = c^{\nu}\,X_{t-1}^{1-\nu}, \qquad \nu \in [0,1]$$

The parameter ν controls the strength of the wealth effect on labor.

Two polar cases.

$\nu = 0$ **GHH** $X_t = X_{t-1}$ is exogenous; consumption and labor enter through $c - v(l)$.
MRS: $w = \psi\, l^{1/\eta}\, X \;\Rightarrow$ **no wealth effect**.

$\nu = 1$ **KPR** $X_t = c$, so disutility scales with consumption.
MRS: $w \propto c\, l^{1/\eta} \;\Rightarrow$ **standard wealth effect, BGP-consistent**.

Budget Constraints

Zero initial and terminal wealth: $a_t^1 = a_t^{61} = 0$.

Worker $(s = 1, \dots, T^W)$

$$a_{t+1}^{s+1} = (1 + r_t)\, a_t^s + (1 - \tau_t)\, w_t\, l_t^s - c_t^s$$

Retiree $(s = T^W + 1, \dots, T)$

$$a_{t+1}^{s+1} = (1 + r_t)\, a_t^s + pen - c_t^s$$

a_t^s	Assets of the s-year-old in period t
r_t	Real interest rate
w_t	Real wage
$\tau_t w_t l_t^s$	Social security (PAYG) contribution
pen	Pension benefit (constant in steady state)

First-Order Conditions

The Lagrangian for the newborn leads to two key conditions:

Intratemporal (labour–leisure trade-off)

$$\frac{u_l(c_t^s, 1 - l_t^s)}{u_c(c_t^s, 1 - l_t^s)} = \gamma \, \frac{c_t^s + \psi}{1 - l_t^s} = (1 - \tau_t)\, w_t$$

The marginal rate of substitution between leisure and consumption equals the after-tax real wage.

Intertemporal (Euler equation)

$$\frac{1}{\beta} = \frac{u_c(c_{t+1}^{s+1}, 1 - l_{t+1}^{s+1})}{u_c(c_t^s, 1 - l_t^s)} \,(1 + r_{t+1}) = \frac{\left(c_{t+1}^{s+1} + \psi\right)^{-\sigma} \left(1 - l_{t+1}^{s+1}\right)^{\gamma(1-\sigma)}}{\left(c_t^s + \psi\right)^{-\sigma} \left(1 - l_t^s\right)^{\gamma(1-\sigma)}} \,(1 + r_{t+1})$$

For retirees set $l_t^s \equiv 0$.

Production Sector

Technology

$$Y_t = F(K_t, L_t) = K_t^{\alpha} L_t^{1-\alpha}$$

Constant returns to scale, perfectly competitive markets.

Profit

$$\Pi_t = Y_t - w_t L_t - (r_t + \delta) K_t$$

Marginal products (factor prices)

$$w_t = (1-\alpha)\, K_t^{\alpha}\, L_t^{-\alpha} \tag{1}$$

$$r_t = \alpha\, K_t^{\alpha-1}\, L_t^{1-\alpha} - \delta \tag{2}$$

α = capital share of income; δ = depreciation rate.

PAYG Social Security & Market Clearing

Government budget (balanced, PAYG)

$$\underbrace{\tau_t\, w_t\, L_t}_{\text{contributions}} = \underbrace{\frac{T^R}{T}\, pen}_{\text{pension outlays}} \qquad \left(\frac{T^R}{T} = \frac{20}{60} \text{ retirees per unit}\right)$$

Capital market clearing

$$A_t \equiv \sum_{s=1}^{T} a_t^s = K_t$$

End-of-period savings equal the aggregate capital stock.

Goods market clearing

$$Y_t = \sum_{s=1}^{T} \frac{c_t^s}{T} + K_{t+1} - (1-\delta)K_t$$

Recursive Competitive Equilibrium

Following Stokey, Lucas and Prescott (1989), the value function of an s-year-old with wealth a^s is $V^s(a^s; K, L, pen)$.

Bellman equation

$$V^s(a^s;\cdot) = \begin{cases} \max\limits_{a^{s+1},c^s,l^s} \left[u(c^s, 1-l^s) + \beta\, V^{s+1}(a^{s+1};\cdot)\right], & s \leq T^W \\ \max\limits_{a^{s+1},c^s} \left[u(c^s, 1) + \beta\, V^{s+1}(a^{s+1};\cdot)\right], & s > T^W \end{cases}$$

An equilibrium is a tuple $\left\{V^s, c^s(\cdot), l^s(\cdot), a^{s+1}(\cdot), w, r\right\}$ such that:

1. Policy rules are individually optimal.
2. Factor prices satisfy (1)–(2).
3. Aggregate consistency: $L = \frac{1}{T}\sum_{s=1}^{T^W} l^s$, $K = \frac{1}{T}\sum_{s=1}^{T} a^s$.
4. Goods and government budgets clear.

Calibration

Calibration

Preferences

Parameter	Value	Target
σ	2.0	IES $= 0.5$
β	0.96	Standard
γ	2.0	$\bar{l} \approx 35\%$
ψ	0.001	Numerical

Technology & Policy

Parameter	Value	Target
α	0.36	Capital share
δ	0.10	Investment ratio
θ	0.30	Replacement rate

Replacement ratio definition

$$\theta = \frac{pen}{(1-\tau)\, w\, \bar{l}} = 0.30, \qquad \bar{l} = \text{average (economy-wide) labour supply}$$

Computation

Inner vs. outer loop: partial vs. general equilibrium

Inner loop — partial equilibrium

Given prices (w, r, τ), solve the household life-cycle:

$$\max \sum_{s=1}^{60} \beta^{s-1} u(c^s, 1 - l^s)$$

s.t. the age-s budget constraints.

Algorithm:

- backward induction $s = 60 \to 1$
- secant / golden-section root of the FOC at each age
- delivers policy functions $\{a^s\}_{s=2}^{60}$, $\{l^s\}_{s=1}^{40}$

Partial: prices treated as *exogenous*.

Outer loop — general equilibrium

Find (K^*, L^*) such that markets clear:

$$K^* = \mathcal{K}(K^*, L^*), \qquad L^* = \mathcal{L}(K^*, L^*)$$

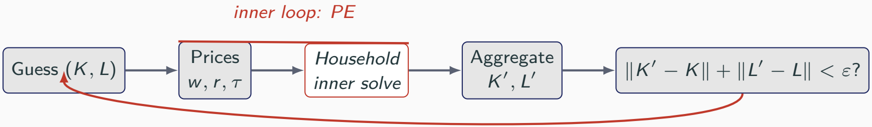
Algorithm: damped Picard iteration

$$(K, L) \leftarrow (1 - \lambda)(K, L) + \lambda\, \mathcal{T}(K, L)$$

with $\mathcal{T} = (\mathcal{K}, \mathcal{L})$ defined by prices $\to$ *inner solve* $\to$ aggregation.

General: prices are *endogenous*; firm FOCs and the government budget must hold *simultaneously* with household optimality.

Inner vs. outer loop: partial vs. general equilibrium



The inner loop closes the household problem *at given prices*; the outer loop closes the *market for capital and labour*. Nesting decouples a hard nonlinear system into two well-posed sub-problems.

Computing the Steady State: Overview

Steady-state conditions:

$$K_t = K_{t+1} = K, \quad L_t = L_{t+1} = L, \quad \{a_t^s\} = \{a^s\}, \quad \{l_t^s\} = \{l^s\}.$$

Fixed-point loop

1. **Guess** $K^{(0)}$ and $L^{(0)}$.
2. **Prices**: compute w, r from (1)–(2); compute τ from the SS budget.
3. **Policy functions**: solve household optimisation for $\{a^s\}_{s=2}^{60}$ and $\{l^s\}_{s=1}^{40}$.
4. **Aggregation**: compute implied K' and L'.
5. **Update**: if $|K' - K| + |L' - L| > \varepsilon$, set $K = (1 - \lambda)K + \lambda K'$ and repeat; else stop.

Step 1 – Initial Guess

Assumption: average labour hours ≈ 0.3

$$L^{(0)} = \frac{T^W}{T} l^{(0)} = \frac{40}{60} \times 0.3 = 0.2$$

Assumption: perfectly smooth utility (flat marginal utility)

Euler equation collapses to $\frac{1}{\beta} = 1 + r$, i.e.

$$r^{(0)} = \frac{1}{\beta} - 1$$

Invert the marginal product of capital:

$$r^{(0)} = \alpha (K^{(0)})^{\alpha-1} (L^{(0)})^{1-\alpha} - \delta \implies K^{(0)} = \left(\frac{\alpha}{r^{(0)} + \delta} \right)^{1/(1-\alpha)} L^{(0)}$$

Step 2 – Prices & Tax Rate

Given K and L:

$$w = (1-\alpha)K^{\alpha}L^{-\alpha}, \qquad r = \alpha K^{\alpha-1}L^{1-\alpha} - \delta.$$

Pension rule: $pen = \theta(1-\tau)w\bar{l}$.

Substituting into the PAYG budget and using $L = \frac{T^W}{T}\bar{l}$:

$$\tau w L = \frac{T^R}{T}\theta(1-\tau)w\bar{l} \implies \tau = \frac{\theta\, T^R/T^W}{1+\theta\, T^R/T^W} = \frac{0.3 \times 0.5}{1 + 0.3 \times 0.5} \approx 0.130.$$

Remark: τ is determined solely by the replacement ratio θ and the demographic ratio T^R/T^W – not by K or L.

Direct Computation

Step 3 — Computing the Policy Function

Two main approaches for solving the household's optimisation problem:

1. Direct computation (AK60_det_direct.ipynb)

Solve the system of non-linear equilibrium equations directly.

+ No income uncertainty, few worker types
+ Fast and accurate when a solution is found
+ **Preferred method if it works**
– Requires a good initial value for the Newton–Raphson algorithm

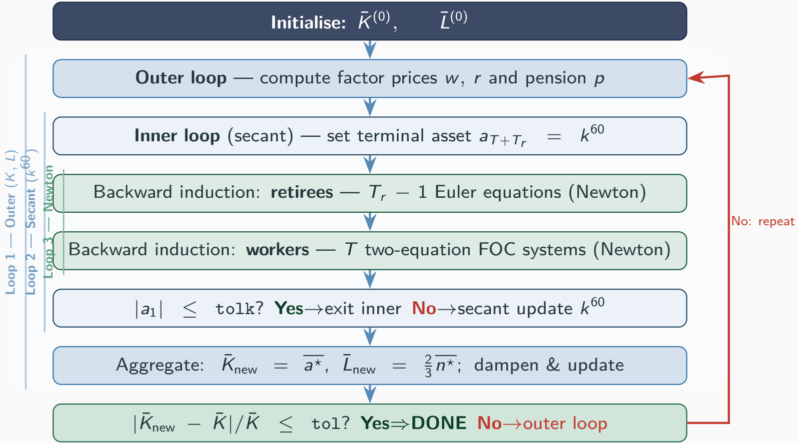
2. Value function iteration (AK60_value_iteration.ipynb)

Solve the Bellman equation recursively on a discretised state space.

+ Handles uncertainty and heterogeneity
– Computationally intensive

→ Often the **only feasible choice** when income risk or agent heterogeneity are present

Full Algorithm: Three Nested Loops



Direct Shooting vs. Value Function Iteration

	Direct Shooting (FOCs)	**Value Function Iteration**
Approach	Solve $\mathcal{R}(x) = 0$	Iterate $V \leftarrow TV$
Solver	Newton / secant	Fixed-point / Howard
State grid	**Not needed**	**Required**
Convergence	Quadratic (local)	Linear
Handles corners	Harder	Easier
Handles non-convexities	No	Yes
Scales to many states	Poorly	Better (EGM)
Code complexity	Moderate	Moderate–High

Direct Computation: Derivation I — Setup

Instantaneous utility

$$u(c, l) = \frac{\left[(c+\psi)(1-l)^{\gamma}\right]^{1-\sigma} - 1}{1-\sigma}, \qquad \tilde{c} \equiv (c+\psi)(1-l)^{\gamma}$$

where $\psi > 0$ ensures $\tilde{c} > 0$ even if $c \to 0$, γ governs the taste for leisure, σ the curvature.

Budget constraints (ψ enters utility only, not the budget)

Workers ($s = 1, \ldots, T^W$):

$$c^s + a^{s+1} = (1+r)\, a^s + (1-\tau)\, w\, l^s$$

Retirees ($s = T^W + 1, \ldots, T$):

$$c^s + a^{s+1} = (1+r)\, a^s + pen$$

with $a^1 = a^{T+1} = 0$.

Direct Computation: Derivation II — Labour FOC

Partial derivatives of u

Let $\tilde{c}^s = (c^s + \psi)(1 - l^s)^\gamma$. Then:

$$\frac{\partial u}{\partial c^s} = (\tilde{c}^s)^{-\sigma}(1 - l^s)^\gamma, \qquad \frac{\partial u}{\partial l^s} = -\gamma\,(\tilde{c}^s)^{-\sigma}(c^s + \psi)(1 - l^s)^{\gamma - 1}$$

Intratemporal FOC: $-\partial u/\partial l^s = (1 - \tau)w \cdot \partial u/\partial c^s$

$$\gamma\,(\tilde{c}^s)^{-\sigma}(c^s + \psi)(1 - l^s)^{\gamma - 1} = (1 - \tau)\,w\,(\tilde{c}^s)^{-\sigma}(1 - l^s)^\gamma$$

Cancelling $(\tilde{c}^s)^{-\sigma}(1 - l^s)^{\gamma - 1}$:

$$\boxed{(1 - \tau)\,w\,(1 - l^s) = \gamma\,(c^s + \psi)}$$

Substituting the budget constraint $c^s = (1 + r)a^s + (1 - \tau)wl^s - a^{s+1}$:

$$(1 - \tau)\,w\,(1 - l^s) = \gamma\left[(1 + r)a^s + (1 - \tau)w\,l^s - a^{s+1} + \psi\right]$$

Direct Computation: Derivation III — Euler Equation (Workers)

Intertemporal FOC with respect to a^{s+1}

$$\frac{\partial u^s}{\partial c^s} = \beta(1+r)\frac{\partial u^{s+1}}{\partial c^{s+1}} \implies \frac{1}{\beta} = (1+r)\frac{\partial u^{s+1}/\partial c^{s+1}}{\partial u^s/\partial c^s}$$

Substituting the marginal utilities

Using $\partial u/\partial c = \tilde{c}^{-\sigma}(1-l)^\gamma$:

$$\frac{\partial u^{s+1}/\partial c^{s+1}}{\partial u^s/\partial c^s} = \frac{(\tilde{c}^{s+1})^{-\sigma}(1-l^{s+1})^\gamma}{(\tilde{c}^s)^{-\sigma}(1-l^s)^\gamma}$$

Expanding $\tilde{c}^s = (c^s+\psi)(1-l^s)^\gamma$ and substituting the budget constraint:

$$\boxed{\frac{1}{\beta} = \frac{\left[(1+r)a^{s+1} + (1-\tau)wl^{s+1} - a^{s+2} + \psi\right]^{-\sigma}(1-l^{s+1})^{\gamma(1-\sigma)}}{\left[(1+r)a^s + (1-\tau)wl^s - a^{s+1} + \psi\right]^{-\sigma}(1-l^s)^{\gamma(1-\sigma)}}(1+r)}$$

Direct Computation: Derivation IV — Retirees & Equation Count

Retiree Euler equation ($l^s = 0$, consumption $= (1+r)a^s + pen - a^{s+1}$)

With $l^s = 0$ the composite consumption simplifies to $\tilde{c}^s = c^s + \psi$, so:

$$\boxed{\frac{1}{\beta} = \frac{\left[(1+r)a^{s+1} + pen - a^{s+2} + \psi\right]^{-\sigma}}{\left[(1+r)a^{s} + pen - a^{s+1} + \psi\right]^{-\sigma}}(1+r)}$$

Counting equations and unknowns

Condition	Equations	Unknowns
Labour FOC, workers $s = 1, \dots, T^W$	40	$\{l^s\}_{s=1}^{40}$: 40
Euler eq., workers $s = 1, \dots, T^W - 1$	39	$\{a^s\}_{s=2}^{T^W+1}$: 40
Euler eq., retirees $s = T^W + 1, \dots, T - 1$	19	$\{a^s\}_{s=T^W+2}^{T}$: 19
Total	**98**	**99**

Plus the terminal condition $a^{T+1} = 0$: **99 nonlinear equations in 99 unknowns**.

Direct Computation: System of Equations

The FOCs are a closed system:

Worker's equations ($s = 1, \ldots, T^W - 1$)

$$(1-\tau)w = \gamma \frac{(1+r)a^s + (1-\tau)w\, l^s - a^{s+1} + \psi}{1 - l^s} \tag{3}$$

$$\frac{1}{\beta} = \frac{\left[(1+r)a^{s+1} + (1-\tau)w\, l^{s+1} - a^{s+2} + \psi\right]^{-\sigma}(1 - l^{s+1})^{\gamma(1-\sigma)}}{\left[(1+r)a^{s} + (1-\tau)w\, l^{s} - a^{s+1} + \psi\right]^{-\sigma}(1 - l^{s})^{\gamma(1-\sigma)}}(1+r) \tag{4}$$

Retiree's equations ($s = T^W + 1, \ldots, T$)

$$\frac{1}{\beta} = \frac{\left[(1+r)a^{s+1} + pen - a^{s+2} + \psi\right]^{-\sigma}}{\left[(1+r)a^{s} + pen - a^{s+1} + \psi\right]^{-\sigma}}(1+r)$$

Reduction to a Single-Variable Problem: The Shooting Idea

Why shooting works

The Euler equations have a **recursive structure**: knowing the asset path from period $s+1$ onward, one can recover a^s by solving one equation backwards. Two boundary conditions pin the problem: $a^1 = 0$ (born with no wealth), $a^{T+1} = 0$ (no bequest).

Algorithm

1. **Guess** $\tilde{a}^T = a^{60}$ (savings in the last period of life).
2. **Retiree backward pass** ($s = T-1, T-2, \ldots, T^W+1$):
   Given (a^{s+1}, a^{s+2}), invert the retiree Euler equation:
$$a^s = \frac{1}{1+r}\left[(\beta(1+r))^{-1/\sigma}(a^{s+1} + pen + \psi) + a^{s+1} - pen\right]$$
3. **Worker backward pass** ($s = T^W, T^W-1, \ldots, 1$):
   Given (a^{s+1}, a^{s+2}), jointly solve the labour FOC and worker Euler equation for (a^s, l^s).
4. **Check:** does the implied $\tilde{a}^1 = 0$?

The Secant Method (Outer Loop)

Root-finding problem

We need to find a^{60} such that $f(a^{60}) = \tilde{a}^1 = 0$.

f has no closed form — it is the output of the backward pass.

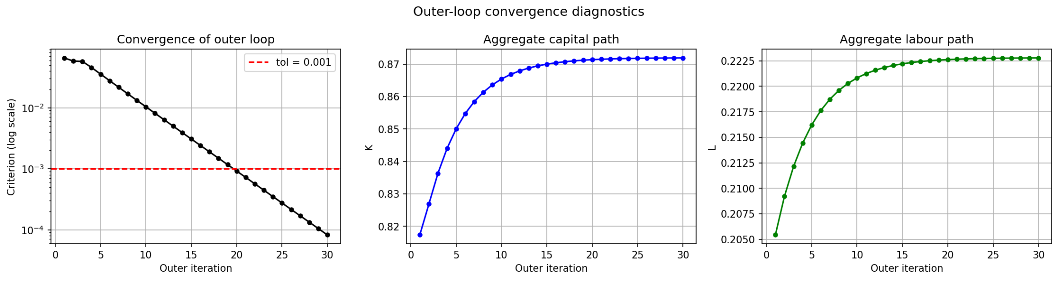
Secant iteration

Start with two guesses x_0, x_1. At each step approximate the derivative by a finite difference and take a Newton-like step:

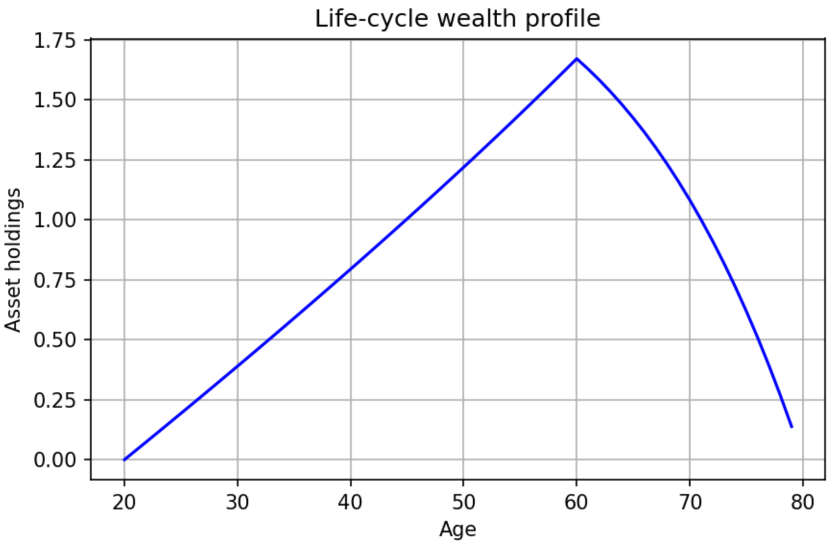
$$x_{n+2} = x_{n+1} - \frac{x_{n+1} - x_n}{f(x_{n+1}) - f(x_n)} f(x_{n+1})$$

- Requires only **two function evaluations to start** (no analytical derivative needed)
- Converges **superlinearly** (order ≈ 1.618)
- Typically ≈ 5 iterations to reach machine precision
- Solution: $a^{60} \approx 0.1382$

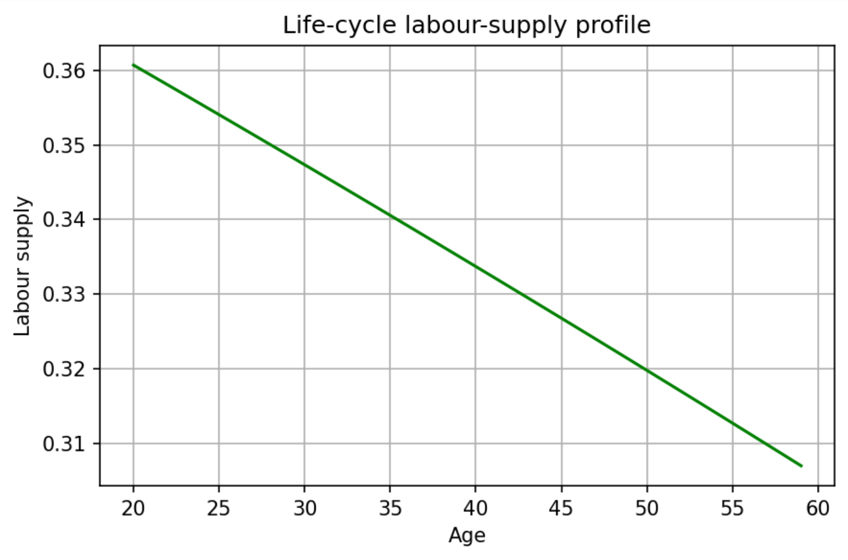
Convegence Diagnostic



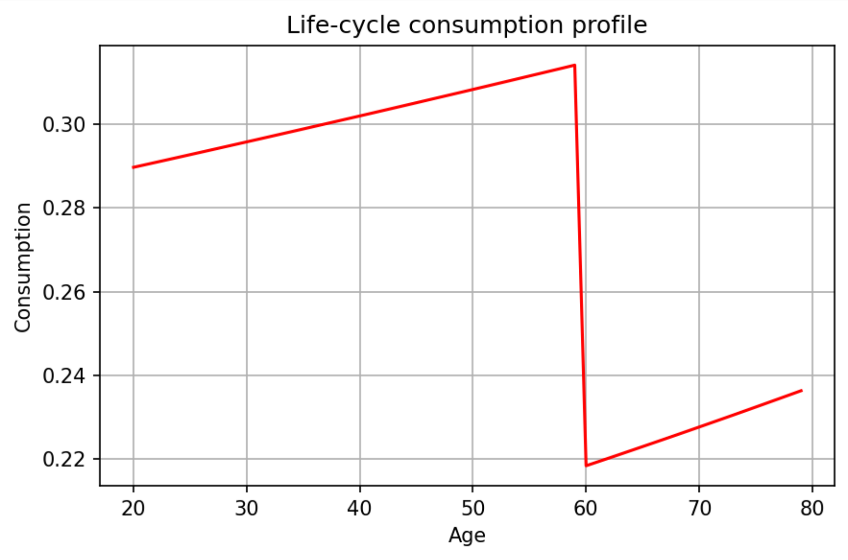
Life-Cycle Profile: Assets



Life-Cycle Profile: Labour Supply



Life-Cycle Profile: Consumption



Empirical Assessment

Model successes

1. Hump-shaped **labour supply** (peaks in mid-40s).
2. Hump-shaped **consumption** – no sharp drop at retirement.

Missing elements (extensions)

- Hump-shaped **individual productivity** profile (empirically peaks $\approx$ age 52).
- **Stochastic survival** (mortality risk).
- **Endogenous retirement** decision.
- **Consumption habits** / durables.

Value Function Iteration

Overview

Three numerical tools to solve finite-horizon dynamic programming problems:

1. **Interpolation** — how to evaluate a function between computed grid points, and why linear interpolation is the standard workhorse.
2. **Golden section search** — how to maximize a unimodal function in one variable without computing derivatives.
3. **Value function iteration (VFI)** — what dynamic programming is, when VFI is the right approach, and how to apply it step by step in the OLG life-cycle model.

1. Why Do We Need Interpolation?

In a numerical model we can only store a function at a *finite* set of points.

The problem. We build a grid $\mathcal{A} = \{a_1, a_2, \dots, a_{n_a}\}$ and compute the value function $V(a_i)$ at each node. But the optimizer will ask for $V(a)$ at arbitrary $a \in [a_{\min}, a_{\max}]$, most of which are *not* on the grid.

Solution: interpolation. Use the known values $\{V(a_i)\}$ to construct a smooth approximation $\tilde{V}$ that agrees with V at all grid points and can be evaluated everywhere else.

Key trade-off

More grid points $\Rightarrow$ higher accuracy but slower computation. The choice of interpolation *scheme* controls how much accuracy we get for a given number of points.

1. Linear Interpolation

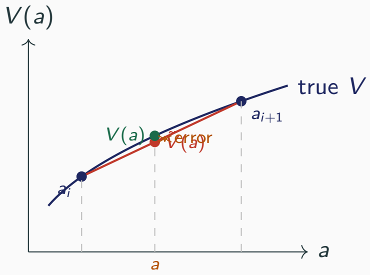
Idea. Between two adjacent grid points a_i and a_{i+1}, approximate V by the straight line connecting $(a_i, V(a_i))$ and $(a_{i+1}, V(a_{i+1}))$.

Define the weight $\lambda = \dfrac{a - a_i}{a_{i+1} - a_i} \in [0, 1]$ and set

$$\tilde{V}(a) \;=\; (1 - \lambda)\, V(a_i) + \lambda\, V(a_{i+1}).$$

Properties:

- Exact at grid nodes ($\lambda = 0$ or 1)
- $O(h^2)$ error where $h = a_{i+1} - a_i$: error halves each time the grid doubles
- Preserves monotonicity
- Very fast: one multiply per evaluation
- Robust to kinks (e.g. binding constraints)



1. Higher-Order Interpolation

Alternatives:

- *Cubic spline* — piecewise cubic polynomials matching.
- *Chebyshev polynomials* — global approximation on $[-1, 1]$ with optimal node placement.
- *Shape-preserving splines* — maintain concavity of the value function.

Why linear is often enough:

- V is smooth and concave in $a \Rightarrow$ small error with 200–500 nodes.
- Higher-order schemes can oscillate near kinks (binding constraints).

Rule of thumb

Use linear interpolation first. Switch to cubic splines or Chebyshev if you need fewer nodes at the same accuracy, or if V is very smooth everywhere.

Warning: Runge's phenomenon

Never use high-degree polynomial interpolation on a uniform grid: oscillations grow at the boundaries and the approximation deteriorates.

In this lecture

Piecewise **linear interpolation** on an equispaced grid with $n_a = 200$ nodes.

The problem. At each grid point a_i and age s we must solve

$$\max_{a' \in [0,\, \bar{a}(a_i, s)]} \Big\{ u\big(c(a_i, a', s),\, l(a_i, a', s)\big) + \beta\, \tilde{V}^{s+1}(a') \Big\},$$

a *one-dimensional* maximization in tomorrow's assets a'.

Why not use calculus?

- $\tilde{V}^{s+1}$ is a piecewise linear function — its derivative has discontinuities at grid nodes.
- Numerical derivatives are noisy and require extra function evaluations.
- Gradient methods (Newton, quasi-Newton) can overshoot or fail near corners of the constraint set.

Key observation. Under standard assumptions the Bellman objective is *strictly concave* (hence *unimodal*) in a'. A *bracketing* method exploiting unimodality is both safe and efficient — it requires no derivatives.

2. Golden Section Search: The Algorithm i

Setting. Maximize a unimodal function f on $[A, D]$.

Step 1. Place two interior probes using the golden ratio $\varphi = \frac{\sqrt{5}-1}{2} \approx 0.618$:

$$B = A + (1 - \varphi)(D - A), \qquad C = A + \varphi(D - A).$$

Step 2. Compare $f(B)$ and $f(C)$:

- $f(B) > f(C)$: maximum lies in $[A, C]$; set $D \leftarrow C$.
- $f(B) < f(C)$: maximum lies in $[B, D]$; set $A \leftarrow B$.

Step 3. In the new interval, *one* old probe is reused; only *one new evaluation* of f is needed.

Step 4. Repeat until $|D - A| < \varepsilon$.

Why φ?

The golden ratio ensures that after each step the surviving probe lands at the golden-ratio position of the new interval, preserving the same geometric structure. This gives the *fastest possible* interval reduction per function evaluation for any bracketing method.

2. Golden Section Search: The Algorithm iii

Convergence. Each iteration reduces the interval by factor $\varphi \approx 0.618$, requiring only **one new function evaluation**. After n iterations:

$$|D_n - A_n| = \varphi^{n-1}(D_0 - A_0).$$

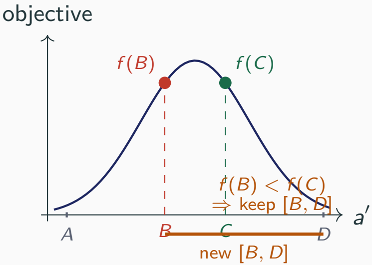
To achieve tolerance ε:

$$n \approx \frac{\ln(\varepsilon/(D_0 - A_0))}{\ln \varphi} \approx 2.08 \log_{10} \frac{D_0 - A_0}{\varepsilon}.$$

Example. Interval $[0, 10]$, tolerance 10^{-6}: $n \approx 43$ evaluations — extremely cheap.

When to use it:

- Function is *unimodal* (one maximum)
- One-dimensional problem



3. Dynamic Programming: The Big Picture

Many multi-period optimization problems can be decomposed into a sequence of smaller, single-period problems. This is **Bellman's Principle of Optimality**:

Principle of Optimality

An optimal policy has the property that, whatever the initial state and initial decision, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.

Implication. Instead of choosing the full sequence $\{c^1, l^1, \ldots, c^T, l^T\}$ simultaneously, solve *backwards*: at each age s, given an optimal continuation value V^{s+1}, choose c^s (and l^s) optimally.

The **value function** $V^s(a^s)$ records the maximum attainable utility from age s onwards given current wealth a^s. It compresses all future optimality conditions into a single sufficient statistic.

3. When Is VFI the Right Tool?

VFI is well suited when:

- Finite horizon (life-cycle) or contraction mapping (infinite horizon, by Blackwell's theorem)
- State space is *low-dimensional* (1–2 continuous states)
- Utility is not differentiable, or constraints are occasionally binding
- *Idiosyncratic uncertainty* in income / health — VFI handles non-linearity and non-normality naturally
- Closed-form Euler equations are unavailable

VFI is *not* the right tool when:

- State space is *high-dimensional*: cost grows as n_a^d where d = number of state variables
- A clean *Euler-equation method* is available and the model has a unique interior solution everywhere
- Very high accuracy is needed on a coarse grid: linear interpolation error may be too large

Curse of dimensionality

Adding one extra state variable multiplies the grid size by n_a, quickly making computation intractable.

The OLG Life-Cycle Model

Demographics. Agents live for $T = 60$ periods: working life $T^W = 40$, retirement $T^R = 20$. No borrowing: $a^1 = a^{61} = 0$.

Lifetime utility of the newborn:

$$\sum_{j=1}^{T} \beta^{j-1} \frac{\left((c^j + \psi)(1 - l^j)^\gamma\right)^{1-\sigma} - 1}{1-\sigma}. \tag{5}$$

Budget constraints:

$$\begin{aligned} a^{j+1} &= (1+r)\, a^j + (1-\tau)\, w\, l^j - c^j, \quad j = 1, \dots, T^W, \\ a^{j+1} &= (1+r)\, a^j + -c^j, \quad j = T^W + 1, \dots, T. \end{aligned}$$

State and controls at age s:

- State: wealth a^s
- Controls: consumption c^s (always), labour supply l^s (workers only)

Value function:

$$V^s(a^s) \equiv \sup_{\mathcal{U}^s(a^s)} \sum_{j=s}^{T} \beta^{s-j} \frac{\left((c^j+\psi)(1-l^j)^\gamma\right)^{1-\sigma}-1}{1-\sigma},$$

where $\mathcal{U}^s(a^s)$ is the set of all feasible control sequences from age s onward. In our application, supremum and maximum coincide.

Terminal condition $(s = 60)$:

Consume everything; no labour; no bequest:

$$c^{60} = pen + (1+r)\,a^{60}, \quad l^{60} = 0, \quad a^{61} = 0, \quad V^{60}(a^{60}) = \frac{\left((c^{60}+\psi)\right)^{1-\sigma}-1}{1-\sigma}.$$

$\Rightarrow$ V^{60} depends only on a^{60}, its state variable — this is the *starting point* of backward induction.

Value Function and Bellman Equations ii

Bellman equation — Retirees ($s = 41, \dots, 59$):

$$V^s(a^s) = \max_{c^s} \frac{\left((c^s + \psi)\right)^{1-\sigma} - 1}{1-\sigma} + \beta\, V^{s+1}(a^{s+1}), \quad \text{s.t.} \;\; a^{s+1} = (1+r)a^s + pen - c^s.$$

Substituting the constraint, this reduces to a *one-dimensional* maximization over $a^{s+1} \in [0, (1+r)a^s + pen]$:

$$\max_{a^{s+1}} \frac{\left(pen + (1+r)a^s - a^{s+1} + \psi\right)^{1-\sigma} - 1}{1-\sigma} + \beta\, V^{s+1}(a^{s+1}).$$

$\Rightarrow$ Solved by **Golden Section Search**.

Value Function and Bellman Equations iii

Bellman equation — Workers ($s = 1, \ldots, 40$):

$$V^s(a^s) = \max_{c^s, l^s} \frac{\left((c^s + \psi)(1 - l^s)^\gamma\right)^{1-\sigma} - 1}{1 - \sigma} + \beta\, V^{s+1}(a^{s+1}), \quad \text{s.t. } a^{s+1} = (1+r)a^s + (1-\tau)wl^s - c^s. \tag{6}$$

Dimension reduction via the labour FOC. With Cobb-Douglas utility, optimal labour satisfies:

$$l^s = \max\left\{0,\ \frac{1}{1+\gamma}\left(1 - \frac{\gamma}{(1-\tau)w}\left(\psi + (1+r)a^s - a^{s+1}\right)\right)\right\}.$$

Substituting, the Bellman reduces to a *one-dimensional* maximization over a^{s+1} — again solved by **Golden Section Search**.

Value Function and Bellman Equations iv

First-order and envelope conditions (interior solution):

If $V^s(\cdot)$ is differentiable, the optimality conditions are:

$$
\begin{aligned}
u_c(c^s, 1 - l^s)\,(1-\tau)w &= u_{1-l}(c^s, l^s) && \text{[intratemporal FOC]} \\
u_c(c^s, 1 - l^s) &= \beta\, V^{s+1\prime}(a^{s+1}) && \text{[Euler equation]} \\
V^{s\prime}(a^s) &= (1+r)\, u_c(c^s, 1 - l^s) && \text{[envelope condition]}
\end{aligned}
$$

It is straightforward to verify that the first equation equals the standard intratemporal labour FOC, and the latter two imply the Euler condition for consumption.

VFI: Grid Setup

Grid. Build a uniform grid for savings:

$$\mathcal{A} = \{a_1, a_2, \ldots, a_{n_a}\}, \quad a_1 = a_{\min} = 0, \quad a_{n_a} = a_{\max} = 10, \quad n_a = 200.$$

$a_{\max} = 10$ is approximately 10 times average wealth in the economy — chosen large enough so that the upper bound never binds in equilibrium.

What we store. For each age $s = 1, \ldots, 60$ and each grid node $i = 1, \ldots, n_a$:

- $V^s(a_i)$ — the value function (a matrix of $60 \times n_a$ numbers)
- $a^{s+1}(a_i)$ — optimal next-period assets (savings policy)
- $c^s(a_i)$ — optimal consumption
- $l^s(a_i)$ — optimal labour supply (workers only)

Off-grid evaluation. Whenever Golden Section Search queries V^{s+1} at a point $a' \notin \mathcal{A}$, we find the bracketing interval $[a_i, a_{i+1}]$ and return the linear interpolant.

VFI: Pseudocode

Algorithm 1 Value Function Iteration — OLG life-cycle model

Require: Grid $\mathcal{A}$; params $\beta, \sigma, \gamma, \psi, r, w, \tau$,
Ensure: $\{V^s,\ a^{s+1}(\cdot),\ c^s(\cdot),\ l^s(\cdot)\}$ for $s = 1, \ldots, T$ Initialise terminal period

for $i = 1, \ldots, n_a$ **do**
  $c_i^T \leftarrow pen + (1+r)\,a_i; \quad V_i^T \leftarrow u(c_i^T, 0)$ consume all; no labour; $a^{T+1} = 0$
end for Backward induction over ages
for $s = T-1, T-2, \ldots, 1$ **do**
  for $i = 1, \ldots, n_a$ **do**
    Set feasible interval $[\underline{a}, \bar{a}]$: $\underline{a} = 0$; $\bar{a}$ from budget constraint
    Golden Section Search on $[\underline{a}, \bar{a}]$: $a_i^{s+1} \leftarrow \arg\max_{a' \in [\underline{a}, \bar{a}]} \left\{ u(c(a_i, a'),\ l(a_i, a')) + \beta\, \tilde{V}^{s+1}(a') \right\}$
      ($\tilde{V}^{s+1}$: piecewise-linear interpolant of $\{V_j^{s+1}\}$)
    Recover c_i^s, l_i^s from a_i^{s+1} via budget constraint / labour FOC
    $V_i^s \leftarrow u(c_i^s, l_i^s) + \beta\, \tilde{V}^{s+1}(a_i^{s+1})$
  end for
end for

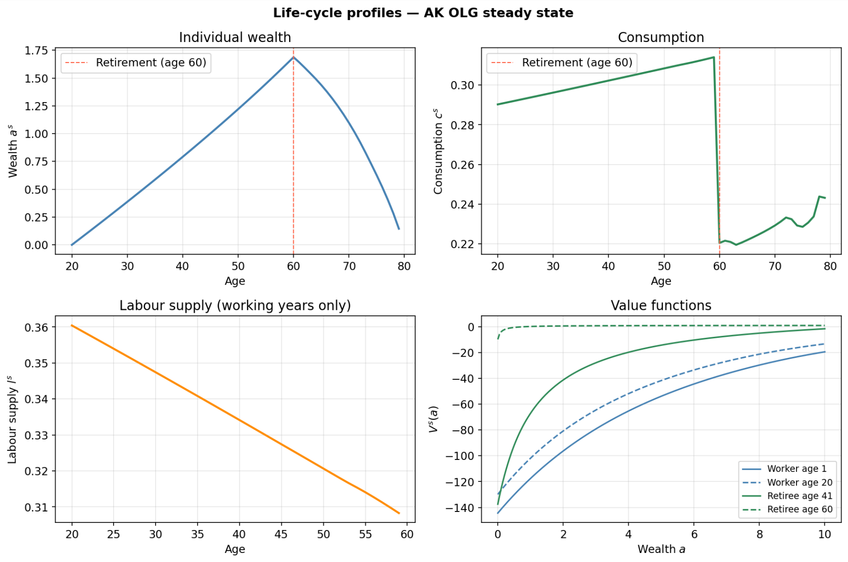
VFI: Step by Step

1. **Build the grid.** $n_a = 200$ equispaced points on $[0, 10]$.
2. **Initialise at $s = 60$.** $c^{60}(a_i) = pen + (1 + r)a_i$; compute $V^{60}(a_i)$.
3. **Solve for retirees, $s = 59, \ldots, 41$.** For each a_i, maximise over a^{s+1} via Golden Section Search, evaluating V^{s+1} by linear interpolation.
4. **Solve for workers, $s = 40, \ldots, 1$.** Same, but the Bellman objective also depends on labour l^s, eliminated via the closed-form FOC. Enforce $l^s \geq 0$.
5. **Save policy and value functions.** Store $\{V^s(a_i),\ a^{s+1}(a_i),\ c^s(a_i),\ l^s(a_i)\}$ for all s and i.
6. **Simulate forward** (separate step). Given $a^1 = 0$ and stored policy functions, trace the individual life-cycle path to produce aggregate statistics and equilibrium objects.

Note on the non-negativity constraint on labour

When the closed-form FOC gives $l^s < 0$, set $l^s = 0$ and re-solve with consumption as the only control.

Life-Cycle Consumption Profile



Life-Cycle Consumption Profile